Experiment 8

Title: Python Program to Find Synonyms and Antonyms of the Word “Active” Using WordNet

---

Aim

To write a **Python program using WordNet** to find synonyms and antonyms of the word *“active”*.

Objectives

1. To understand the use of WordNet in Python for semantic analysis.
2. To identify synonyms and antonyms of a word using WordNet.
3. To implement a Python program for extracting word relationships.

Theory

1. Introduction to WordNet in Python
WordNet is a **lexical database for the English language** that groups words into sets of synonyms called synsets. In Python, WordNet is available through the **Natural Language Toolkit (NLTK)** library. It helps in analyzing the meanings and relationships between words.

2. Synonyms
Synonyms are words that **have the same or similar meanings**. WordNet organizes synonyms into synsets, where each synset represents a specific concept. Using WordNet functions in Python, we can retrieve synonyms for a given word easily.

3. Antonyms
Antonyms are words that **have opposite meanings**. WordNet provides antonyms through relationships associated with words in a synset. By accessing these relationships using NLTK, it is possible to find opposite words for a given term.

4. Applications of WordNet in NLP
WordNet is widely used in **text analysis, search engines, machine translation, and question answering systems**. It helps improve language understanding by identifying relationships between words and their meanings.

Conclusion

In this experiment, a Python program was written to find synonyms and antonyms of the word *"active"* using WordNet. WordNet provides a structured database that helps identify relationships between words. Using the NLTK library, these relationships can be extracted easily in Python. This method helps improve semantic understanding of text. Therefore, WordNet is an important tool in natural language processing tasks.

Expt 9 case study
Expt 10 Ma'am will confirm
Expt 11 Lab Project